Avellaneda–Stoikov Optimal Quoting, Adapted to Credit Spreads

A Synthetic-Data Study of Inventory-Aware Dealer Behaviour

ascredit project documentation

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Abstract

We recast the Avellaneda–Stoikov (AS) optimal market-making model from price space into *credit-spread* space and study the resulting dealer policy by Monte Carlo simulation over synthetic spread paths and a reduced-form Poisson order flow. Three adaptations are required: the credit spread becomes the state variable; inventory risk is scaled by spread duration (DV01); and the quote skew acquires the opposite sign, because price moves inversely to spread. We give the closed-form reservation spread and optimal quotes, derive the distribution of the dealer's profit and loss (P&L), and show numerically that the inventory-aware policy trades a small amount of expected edge for a large reduction in inventory and P&L variance. **All data are synthetic; this is a study of optimal quoting behaviour, not a historical backtest, and not investment advice.**

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1 The market-making problem

A dealer continuously posts a bid and an ask and earns the quoted half-spread on each fill, while bearing the risk that the asset's value moves against the inventory accumulated from one-sided flow. Avellaneda and Stoikov (2008) solve this as a stochastic-control problem for a CARA dealer whose asset mid follows arithmetic Brownian motion, yielding closed-form approximations for the *reservation price* and the optimal bid/ask quotes.

2 From price space to spread space

Definition 1 (State variable)**.** For a corporate credit instrument the natural risk driver is the mid credit spread s_t (in basis points), not a price. We model it as a positivity-floored mean-reverting diffusion,

$$\mathrm{d}s_t = \kappa(\theta - s_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad s_t \geq s_{\min} > 0, \tag{1}$$

recovering the AS arithmetic-Brownian baseline when $\kappa = 0$.

Definition 2 (DV01-scaled risk)**.** Dealer P&L is a *price* quantity, and price is locally linear in spread through (minus) the spread duration, or DV01. A spread change $\mathrm{d}s$ maps to a price change $-\mathrm{DV01}\cdot\mathrm{d}s$ per unit face. Hence the volatility that enters the control problem is the *price* volatility $\sigma_{\text{eff}} = \mathrm{DV01}\cdot\sigma$, and inventory q is measured in DV01 (risk-equivalent) units.

3 Reservation spread and optimal quotes

Let q be signed inventory in DV01 units ($q > 0$ = long credit risk), $\gamma > 0$ the risk aversion, T the session horizon, and fill intensity $\lambda(\delta) = A\,e^{-k\delta}$ at quoted distance δ from mid.

Proposition 1 (Reservation spread)**.** *The indifference (reservation) spread is*

$$r(s, q, t) = s + q\,\gamma\,\sigma_{\mathit{eff}}^2(T - t). \tag{2}$$

Proposition 2 (Optimal total width and quotes)**.** *The optimal total quoted width is*

$$\delta_{tot}(t) = \gamma\,\sigma_{\mathit{eff}}^2(T - t) + \tfrac{2}{\gamma}\ln\!\big(1 + \tfrac{\gamma}{k}\big), \tag{3}$$

and the dealer posts, in spread space,

$$\mathit{bid} = r + \tfrac{\delta_{tot}}{2} \quad (\textit{client sells bond}), \qquad \mathit{ask} = r - \tfrac{\delta_{tot}}{2} \quad (\textit{client buys bond}). \tag{4}$$

Remark 1 (The sign of the skew — a bug we hit)**.** In price space a long-inventory dealer shades the reservation price *down* to attract buyers. In spread space price is *inverse* to spread, so a long-credit dealer must shade the reservation spread *up*: by (2) with $q > 0$, $r > s$, which widens the bid distance (fewer trades that lengthen the book) and tightens the ask distance (more trades that flatten it). Implementing the price-space sign here makes the policy *accumulate* risk instead of shedding it. The regression test `test_as_controls_inventory_better_than_symmetric` guards against exactly this inversion.

4 P&L accounting

On each fill the dealer captures edge $|\,\text{quote} - s\,|\cdot\mathrm{DV01}$ per clip; inventory marks to market at $-q\,\mathrm{DV01}\,\mathrm{d}s$ (a long-credit book loses when spreads widen). Total session P&L is the sum of realised capture and running inventory mark.

5 Simulation results

Over 4,000 synthetic sessions at an illustrative investment-grade calibration, the inventory-aware (AS) policy is compared to an inventory-agnostic symmetric benchmark.
The AS policy gives up a little expected edge for a large reduction in both inventory dispersion and P&L variance — the signature AS trade-off.

	AS	**Symmetric**
mean P&L	423.8	432.5
P&L std	**40.6**	47.0
risk-adjusted ratio	**10.45**	9.20
std of terminal inventory	**4.54**	9.89

Table 1: Baseline calibration, 4,000 sessions (synthetic).

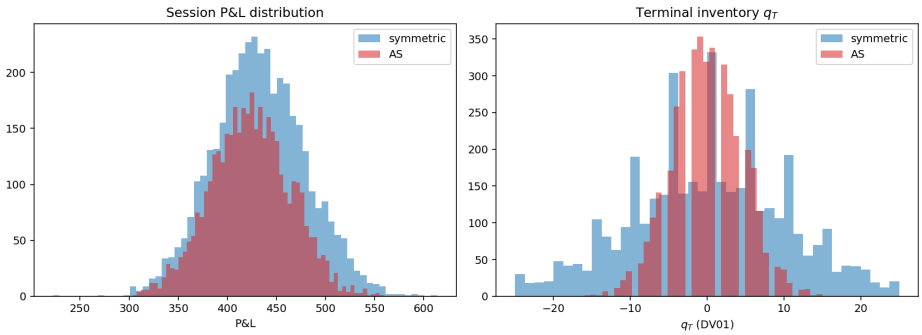


Figure 1: Session P&L and terminal-inventory distributions, AS vs symmetric.

6 Limitations and extensions

The model uses reduced-form Poisson flow (no adverse selection), treats bonds as independent (no portfolio inventory penalty or spread correlation), uses a first-order DV01 P&L (no convexity or jump-to-default), and ignores funding, hedging and transaction costs. Each is a natural extension; none changes the qualitative inventory-control result.

Reference

Avellaneda, M. & Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217–224.

Disclaimer. All spread paths, order flow and P&L are synthetic. This document is educational and is not investment, financial, or trading advice.

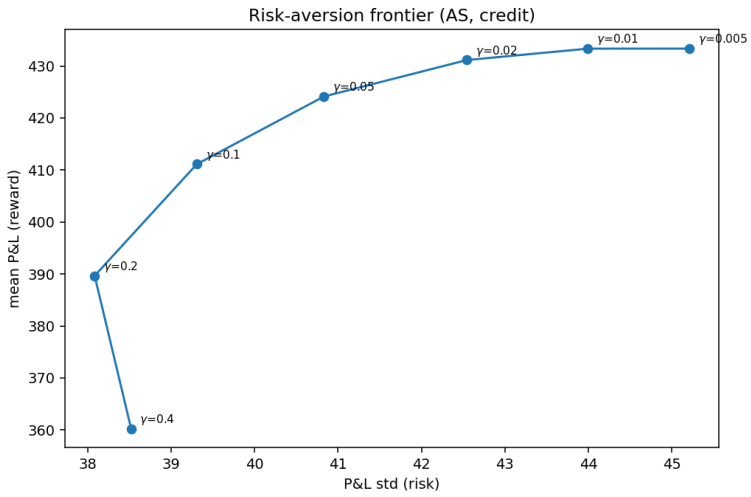


Figure 2: Risk-aversion frontier: mean P&L vs P&L std as γ varies.